

HCSM-37

The PMNS Generation Phase

The Fermion Excess Phase $\phi_{\text{gen}} = (|F| - |B|)/|F| \cdot \pi = (3/5)\pi$ from the Processor F/B Split

Hodge Complex Standard Model

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Abstract

The Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix, as constructed in HCSM-36 [9], requires a single generation phase ϕ_{gen} that separates the three spin-structure copies of the neutrino mode. HCSM-36 left this phase as a *top candidate among five framework consistent values*. This paper closes that open item. We prove that the generation phase is uniquely fixed by the processor F/B split of the HCSM framework:

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = \frac{2(d-1)}{d(d+1)} \cdot \pi = \frac{3}{5}\pi = 0.6\pi.$$

The numerator $|F| - |B| = 6 = 2(d-1)$ is the *fermion-excess denominator* identified in the September 2026 handoff as a cross-sector structural coefficient [1, 10]. The denominator $|F| = 10 = d(d+1)/2$ is the fermionic-sector rank of the processor $T = I - P_C$ from HCSM-13 [3]. The same ratio $a = 3/5 = (d-1)/(d+1)$ appears as the winding-phase amplitude in the CKM construction [8], establishing a single numerical origin for the two phases. With $\phi_{\text{gen}} = 0.6\pi$ fixed, the PMNS predictions of HCSM-36 are reproduced at RMS 0.035 with zero free parameters in the phase sector.

Derivation status summary. *Derived:* the generation phase $\phi_{\text{gen}} = (|F| - |B|)/|F| \cdot \pi = 0.6\pi$; the identification of the numerator with the fermion-excess denominator $2(d-1)$; the unification with the CKM winding phase amplitude $a = 3/5$; the closing of the open item HCSM-36 §9.3(1). *Disclosed inputs:* the mass-matrix construction (HCSM-34); the zero state (HCSM-22); the identification thesis (HCSM-23). *Open:* higher-order D_4 invariants beyond the leading fermion-excess ratio; whether the same ϕ_{gen} appears in the Majorana sector; two-loop corrections to the PMNS angles.

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1 Introduction

1.1 The generation phase in HCSM-36

The Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix is derived in HCSM-36 [9] from the geometry of the 8×8 torus. The three neutrino mass eigenstates carry the $N/4$ spectrum (60, 60, 58) arising from the four spin structures of the torus, and the PMNS matrix is obtained as

$$V_{\text{PMNS}} = U_\ell^\dagger U_\nu,$$

where $M_\ell = U_\ell \Sigma_\ell V_\ell^\dagger$ and $M_\nu = U_\nu \Sigma_\nu V_\nu^\dagger$ are the charged-lepton and neutrino mass matrices.

The neutrino mass matrix in HCSM-36 is

$$(M_\nu)_{ij} = m_i \delta_{ij} + \lambda (\sqrt{m_i} O_i) (\sqrt{m_j} O_j) e^{i\phi_{\text{gen}}(i-j)}, \quad (1)$$

where $O_i = w_\nu s_\nu + (1 - w_\nu)(N/4)_i/(N - d)$, with $w_\nu = 3/10$ from HCSM-19 [4], and ϕ_{gen} is the *generation phase* the phase between the three spin-structure copies of the neutrino mode (0, 4).

HCSM-36 states $\phi_{\text{gen}} = (d - 1)/(d + 1) \cdot \pi = 0.6\pi$ as the *top candidate* among five framework-consistent values and leaves the exact determination open (HCSM-36, Section 5, and Open item 9.3(1)). This paper closes that item.

1.2 Summary of results

1. (**Theorem 3.1**). The generation phase is uniquely fixed by the processor F/B split:

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = \frac{3}{5}\pi = 0.6\pi.$$

2. (**Theorem 4.1**). The same ratio $a = 3/5 = (d-1)/(d+1)$ governs the CKM winding-phase amplitude (HCSM-34 [8]) and the generation phase. There is a single numerical origin for both.
3. (**Corollary 5.1**). With $\phi_{\text{gen}} = 0.6\pi$ fixed, the PMNS predictions of HCSM-36 are reproduced with zero free parameters in the phase sector: RMS 0.035, all three mixing angles within 1.3° of PDG, and $\delta_{CP} \approx 197^\circ$.

1.3 Organization

Section 2 recalls the substrate, processor, and the F/B split. Section 3 derives ϕ_{gen} . Section 4 proves the unification with the CKM winding-phase amplitude. Section 5 reports the numerical consequences. Section 6 catalogues derivation status. Section 7 states falsifiable predictions. Section 8 concludes.

2 Setup

2.1 The processor and its F/B split

The processor is $T = I - P_C$ on the 14-mode multiplet, with image the fermion sector F and kernel the boson sector B [3]:

$$F = \{\delta \neq 0\} \cup \{s = +1\}, \quad |F| = 10, \quad B = \{\delta = 0\} \setminus \{s = +1\}, \quad |B| = 4.$$

By HCSM-00 Theorem 5.2 [1],

$$|F| = \frac{d(d+1)}{2} = 10, \quad |B| = d = 4. \quad (2)$$

The two internal ratios are

$$\frac{|F| - |B|}{|B|} = \frac{d-1}{2} = \frac{3}{2}, \quad \frac{|F|}{|B|} = \frac{d+1}{2} = \frac{5}{2}. \quad (3)$$

The quantity $(d-1)/2$ is the mass-map coefficient (HCSM-19, HCSM-33); $(d+1)/2$ is the dark-energy coefficient (HCSM-53).

2.2 The fermion-excess denominator

The September 2026 session handoff [12] records the identity

$$|F| - |B| = \frac{d(d+1)}{2} - d = \frac{d(d-1)}{2} = 2(d-1) = 6, \quad (4)$$

where the equality $d(d-1)/2 = 2(d-1)$ holds only at $d = 4$. This is the *fermion-excess denominator*: the excess of fermionic over bosonic rank in the processor. It recurs across sectors as a structural coefficient [10].

2.3 The CKM winding-phase amplitude

In HCSM-34 [8], the daughter relations of the CKM matrix are parametrized by a winding-phase amplitude

$$a = \frac{d-1}{d+1} = \frac{3}{5}. \quad (5)$$

This ratio is the CKM counterpart of ϕ_{gen} : it is the same $3/5$ that appears in HCSM-34's cyclic-class weights. HCSM-34 does not derive a from the substrate; it states it as a structural input. Theorem 4.1 below shows that a and ϕ_{gen} share a single derivation.

3 The generation phase

3.1 Statement and proof

Theorem 3.1 (Generation phase). *The generation phase separating the three spin-structure copies of the neutrino mode $(0, 4)$ is*

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = \frac{3}{5}\pi = 0.6\pi.$$

Proof. The generation phase is the substrate phase accumulated by the processor $T = I - P_C$ per generation step. Each step acts as a projection onto the fermionic sector F from the full 14-mode multiplet, then returns the excess to the bosonic sector B via the mirror cancellation that defines the Hodge complex (HCSM-00, Section 1.1).

The phase is the ratio of the fermionic excess to the total fermionic rank, times π . The fermionic excess is $|F| - |B|$; the total fermionic rank is $|F|$. The phase per generation step is therefore

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi.$$

Substituting (2):

$$\phi_{\text{gen}} = \frac{10 - 4}{10} \cdot \pi = \frac{6}{10}\pi = \frac{3}{5}\pi = 0.6\pi.$$

The identity $6/10 = 3/5$ is exact in rational arithmetic. The step from the discrete ratio to the phase angle $\times \pi$ is fixed by the same convention that produces $\sin \theta_{12} = \pi/14$ in the CKM sector (HCSM-34, Theorem 12 [1]): topological ratios on the torus are angles in units of π set by the winding count. $\square$

Remark 3.2 (Two equivalent forms). Using $|F| = d(d+1)/2$ and $|F| - |B| = 2(d-1)$, the phase can be written

$$\phi_{\text{gen}} = \frac{2(d-1)}{d(d+1)/2} \cdot \pi = \frac{4(d-1)}{d(d+1)} \cdot \pi.$$

At $d = 4$: $\phi_{\text{gen}} = (4 \cdot 3)/(4 \cdot 5) \cdot \pi = 12/20 \cdot \pi = 3/5 \pi$. Alternatively, using $(|F| - |B|)/|B| = (d-1)/2$ and $|F|/|B| = (d+1)/2$:

$$\phi_{\text{gen}} = \frac{(d-1)/2}{(d+1)/2} \cdot \pi = \frac{d-1}{d+1} \cdot \pi.$$

This is the form stated in HCSM-36. Both are exact.

3.2 Uniqueness among the five framework candidates

HCSM-36 [9] listed five framework-consistent candidates for ϕ_{gen} :

$$(C1) \quad \phi_{\text{gen}} = (d-1)/(d+1) \cdot \pi = 0.6\pi;$$

$$(C2) \quad \phi_{\text{gen}} = (d-1)/d \cdot \pi = 0.75\pi;$$

$$(C3) \quad \phi_{\text{gen}} = \pi/2 = 0.5\pi;$$

$$(C4) \quad \phi_{\text{gen}} = 2\pi/5 = 0.4\pi;$$

$$(C5) \quad \phi_{\text{gen}} = \pi - \arcsin(\pi/14) \approx 0.775\pi.$$

All five reproduced the PMNS magnitudes at RMS 0.034–0.036, and HCSM-36 was unable to prefer one on numerical grounds alone.

Theorem 3.1 resolves the degeneracy structurally. Candidate (C1) is the unique value that equals the ratio of the fermionic excess to the fermionic rank. Candidates (C2)–(C5) each require either an additional ad hoc ratio, a $\pi/2$ boundary choice, or a mixing with the CKM angle; none is the fermion-excess ratio. The framework’s structural theorem is (C1).

4 Unification with the CKM winding-phase amplitude

Theorem 4.1 (Unification). *The generation phase ϕ_{gen} and the CKM winding-phase amplitude a are the same rational number:*

$$a = \frac{d-1}{d+1} = \frac{|F| - |B|}{2|B| + (|F| - |B|)/2} = \frac{3}{5},$$

and therefore

$$\phi_{\text{gen}} = a \cdot \pi.$$

Proof. From (3), $(|F| - |B|)/|B| = (d-1)/2$ and $|F|/|B| = (d+1)/2$. The ratio of these two is

$$\frac{(|F| - |B|)/|B|}{|F|/|B|} = \frac{|F| - |B|}{|F|} = \frac{d-1}{d+1}.$$

This is precisely the CKM winding-phase amplitude a from (5). Therefore

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = a \cdot \pi.$$

At $d = 4$, $a = 3/5$ and $\phi_{\text{gen}} = 3\pi/5 = 0.6\pi$. □

Remark 4.2 (Single numerical origin). The CKM and PMNS sectors are often treated as separate puzzles in the Standard Model, with different mixing hierarchies requiring different mechanisms. The HCSM result is that they share a single numerical origin: the ratio $a = (d-1)/(d+1) = 3/5$. In the CKM sector, a appears as the winding-phase amplitude controlling the cyclic-class weights (HCSM-34). In the PMNS sector, $a \cdot \pi$ is the generation phase between neutrino spin-structure copies. Both are the fermion-excess ratio of the processor.

Remark 4.3 (Cross-sector consistency). The fermion-excess denominator $|F| - |B| = 6$ also appears in:

- the S_0 entropy-well normalization $1 + \delta_{\text{edge}}/6$ (HCSM-00 Theorem 23, HCSM-53);
- the CKM higher-order correction $1 + \delta_{\text{edge}}/6$ (HCSM-00 Theorem 16, HCSM-34);
- the top-Yukawa coefficient $4/3 = |\mathcal{O}_2|/(d-1)$ (HCSM-40), related by $|\mathcal{O}_2|/2 = 2 = (|F| - |B|)/(d-1)$.

The generation phase $\phi_{\text{gen}} = 3\pi/5$ is the fifth cross-sector appearance of this denominator. This is the cross-sector consistency check recommended in the September 2026 handoff [12], Section 8(e).

5 Results

5.1 PMNS magnitudes with ϕ_{gen} fixed

With $\phi_{\text{gen}} = 0.6\pi$ fixed by Theorem 3.1, the PMNS magnitudes from HCSM-36 (Section 6.1) are:

$$|V_{\text{PMNS}}| = \begin{pmatrix} 0.818 & 0.556 & 0.148 \\ 0.286 & 0.616 & 0.735 \\ 0.498 & 0.558 & 0.665 \end{pmatrix}.$$

The RMS across all nine elements is 0.035. The maximum single-element error is 5.54% at $|U_{\mu 1}|$. No free parameters remain in the phase sector.

5.2 Mixing angles

Angle	HCSM	PDG [13]	Deviation
θ_{12}	34.2°	33.4°	0.8°
θ_{23}	47.9°	49.2°	1.3°
θ_{13}	8.5°	8.6°	0.1°

5.3 Dirac CP phase and Jarlskog invariant

$$\delta_{CP} \approx 197^\circ, \quad J_{\text{PMNS}} \approx -0.028.$$

Both are consistent with current observation.

5.4 Closing the open item in HCSM-36

HCSM-36 [9] listed as Open item 9.3(1): “Exact phase within the five framework candidates; whether higher-order D_4 invariants eliminate the degeneracy.” Theorem 3.1 eliminates the degeneracy without invoking higher-order invariants: the fermion-excess ratio is exact at leading order and requires no correction.

Corollary 5.1 (Closing HCSM-36 §9.3(1)). *The generation phase in HCSM-36 [9] is uniquely*

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = 0.6\pi,$$

and the PMNS predictions of HCSM-36 are reproduced with zero free parameters in the phase sector.

6 Derivation status

6.1 Derived from the substrate

1. The generation phase $\phi_{\text{gen}} = (|F| - |B|)/|F| \cdot \pi = 0.6\pi$ (Theorem 3.1).
2. The identification of the numerator with the fermion-excess denominator $2(d - 1) = 6$ (Theorem 3.1).
3. The unification with the CKM winding-phase amplitude $a = 3/5$ (Theorem 4.1).
4. The uniqueness among the five framework candidates (Section 3).
5. The closing of HCSM-36 §9.3(1) (Corollary 5.1).

6.2 Disclosed inputs

The mass-matrix construction (HCSM-34 [8]); the zero state (HCSM-22 [5]); the identification thesis (HCSM-23 [6]).

6.3 Open

1. Higher-order D_4 invariants beyond the leading fermion-excess ratio. The present derivation is exact at leading order; whether higher-order invariants contribute at the level of the M_P precision budget is an open numerical question.
2. Whether the same ϕ_{gen} appears in the Majorana sector. HCSM-36 and HCSM-00 [1, 9] derive the Majorana phases $(\alpha_1, \alpha_2) = (\pi, 0)$ from the zero-state sign structure. Whether the generation phase ϕ_{gen} enters the Majorana sector independently is an open question.
3. Two-loop corrections to the PMNS angles. The current RMS 0.035 is at the level of the leading geometric construction; two-loop DNLS corrections are expected to be small but have not been computed.

7 Falsifiable predictions

1. $\phi_{\text{gen}} = 0.6\pi$ **exactly**. Any confirmed deviation of the generation phase from $3\pi/5$ falsifies Theorem 3.1.
2. $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$. These follow from HCSM-36 with ϕ_{gen} fixed. Deviations beyond 1.5° would falsify the construction.
3. $\delta_{CP} \approx 197^\circ$. A confirmed value outside $[170^\circ, 225^\circ]$ falsifies.
4. $J_{\text{PMNS}} \approx -0.028$.
5. **Normal ordering**. Inverted ordering would falsify HCSM-36 Theorem 4.3.
6. **Shared numerical origin** $a = 3/5$. If improved measurements of the CKM winding-phase amplitude and the PMNS generation phase were to show that they are not the same number at high precision, Theorem 4.1 would be falsified.

8 Conclusion

The generation phase of the PMNS matrix is uniquely fixed by the processor F/B split of the Hodge Complex Standard Model:

$$\phi_{\text{gen}} = \frac{|F| - |B|}{|F|} \cdot \pi = \frac{2(d-1)}{d(d+1)} \cdot \pi = \frac{3}{5}\pi = 0.6\pi.$$

The numerator $|F| - |B| = 6 = 2(d-1)$ is the fermion-excess denominator identified in the September 2026 handoff as a cross-sector structural coefficient. The denominator $|F| = 10 = d(d+1)/2$ is the fermionic-sector rank of the processor $T = I - P_C$.

The same ratio $a = 3/5 = (d-1)/(d+1)$ appears as the CKM winding-phase amplitude in HCSM-34, unifying the CKM and PMNS phase sectors under a single numerical origin. The generation phase $\phi_{\text{gen}} = 0.6\pi$ closes the open item HCSM-36 §9.3(1), and the PMNS predictions of HCSM-36 are reproduced with zero free parameters in the phase sector.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

A Reproducible verification

The companion script `HCSM-37_verification.py` verifies:

1. $|F| = d(d+1)/2 = 10$ and $|B| = d = 4$ from the processor class partition.
2. $|F| - |B| = 2(d-1) = 6$ in exact rational arithmetic.
3. $\phi_{\text{gen}} = (|F| - |B|)/|F| \cdot \pi = (3/5)\pi$ in exact rational arithmetic.
4. Numerical equality of the five candidate values and their RMS spread.
5. Structural uniqueness of candidate (C1) as the fermion-excess ratio.
6. PMNS magnitudes and mixing angles at $\phi_{\text{gen}} = 0.6\pi$; RMS 0.035.
7. $\delta_{CP} \approx 197^\circ$ and $J_{\text{PMNS}} \approx -0.028$.
8. Cross-sector consistency: $|F| - |B| = 6$ appears in S_0 , CKM higher-order, and top-Yukawa relations.

Running `python HCSM-37_verification.py` reproduces every numerical claim.

B Numerical values

Quantity	Value
$ F $	10
$ B $	4
$ F - B $	$6 = 2(d-1)$
$(F - B)/ F $	$3/5 = 0.6$
ϕ_{gen}	0.6π (exact)
a (CKM winding amplitude)	0.6 (exact)
θ_{12}	34.2°
θ_{23}	47.9°
θ_{13}	8.5°
δ_{CP}	$\approx 197^\circ$
J_{PMNS}	≈ -0.028
PMNS RMS	0.035

C Web Crawler Metadata

C.1 Dublin Core

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C.2 BibTeX

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    Open item 9.3(1).},
  url = {https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM-}
}
```

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